Combined Oxidative Phosphorylation Deficiency Type 3 (COXPD3) — Comprehensive Disease Report

Causal gene: *TSFM* (mitochondrial translation elongation factor Ts, EF-Ts). COXPD3 is an autosomal-recessive mitochondrial disease caused by biallelic pathogenic variants in the nuclear gene **TSFM**, which encodes mitochondrial elongation factor Ts (EF-Ts). EF-Ts is the guanine-nucleotide-exchange factor (GEF) for elongation factor Tu (EF-Tu/TUFM); loss of EF-Ts destabilizes EF-Tu and cripples synthesis of the 13 mtDNA-encoded oxidative-phosphorylation (OXPHOS) subunits, producing a **combined (multiple) respiratory-chain complex deficiency** — the defining biochemical lesion of this disorder (17033963).

> Evidence source types are flagged throughout as **[human clinical]**, **[in vitro]**, **[computational]**, or **[model organism]**.

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1. Disease Information

Overview. COXPD3 is a rare, severe, clinically heterogeneous primary mitochondrial disorder of mitochondrial protein synthesis. Because the 13 polypeptides encoded by mtDNA are all core OXPHOS subunits, a translation-elongation defect reduces assembly of complexes I, III, IV and V simultaneously (complex II, entirely nuclear-encoded, is spared), impairing ATP generation in energy-dependent tissues (heart, brain, optic nerve, liver, skeletal muscle). **[human clinical / in vitro]** (17033963)

Key identifiers. - **MONDO:** MONDO:0012512 - **OMIM (phenotype):** 610505 (COXPD3); **OMIM (gene):** 604723 (TSFM) - **Orphanet:** mitochondrial DNA translation-defect / combined OXPHOS deficiency group (ORPHA:254913 "Combined oxidative phosphorylation defect"; subtype-level)

- **ICD-11:** 5C53.0 (disorders of mitochondrial energy metabolism / nuclear-encoded) — code at category level - **ICD-10:** E88.49 (other mitochondrial metabolism disorders) - **MeSH:** "Mitochondrial Diseases"; "OXPHOS" / "Oxidative Phosphorylation" (no COXPD3-specific MeSH) - **Gene:** HGNC:11701, NCBI Gene 10102, Ensembl ENSG00000123297, UniProt P43897

Synonyms: combined oxidative phosphorylation deficiency type 3; TSFM combined oxidative phosphorylation deficiency; combined oxidative phosphorylation deficiency caused by mutation in TSFM; fatal mitochondrial disease due to COXPD3; (EF-Ts deficiency; mitochondrial translation elongation factor Ts deficiency).

Information source. Derived from **aggregated disease-level resources** (OMIM, Orphanet) built from **individual patient case reports and small case series** — this is an ultra-rare disorder with only a few dozen reported families; no large EHR cohorts exist.

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2. Etiology

Causal factors — genetic (Mendelian). Biallelic (homozygous or compound-heterozygous) loss-of-function/hypomorphic variants in **TSFM**. No environmental or infectious cause; disease is fully determined by genotype. **[human clinical]** (P 17033963 P 21741925 P 25037205).

Genetic risk factors. - Causal variants: see Section 4. The recurrent **c.997C>T (p.Arg333Trp;** older numbering **R312W)** is the most frequently reported allele and causes the most severe outcomes. - **Consanguinity** underlies many homozygous cases (reported Pakistani, Mediterranean, and other families). - **Founder effect:** Finnish p.Gln286X (c.856C>T) nonsense allele with carrier frequency ~1:80 (P 25037205).

Environmental / lifestyle / infectious factors. None established as causal. As in other mitochondrial diseases, **intercurrent infection, fever, fasting, and catabolic/metabolic stress** can precipitate acute decompensation (metabolic crises, lactic acidosis) — a gene-environment interaction at the level of energy demand rather than disease causation.

Certain mitochondrial toxins (e.g., aminoglycosides, valproate, some anaesthetics) are best avoided but are not disease triggers per se. **[inferred from mitochondrial-disease class knowledge]**

Protective factors. None genetically defined. The compensatory up-regulation of EF-Tu and mitochondrial biogenesis seen in some patient tissues is an endogenous buffering mechanism that may mitigate severity in sparable tissues (P 30911037) **[in vitro]**.

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3. Phenotypes

COXPD3 shows **marked phenotypic heterogeneity**; the same genotype (R333W) can present as encephalomyopathy, cardiomyopathy, or liver failure (P 21741925).

Core manifestations with suggested HPO terms:

Phenotype	Type	Onset	Severity / course	Frequency	HPO
Hypertrophic cardiomyopathy	clinical sign	neonatal–infantile	severe, often progressive	common (a presenting feature in many)	HP:0001639
Dilated cardiomyopathy (± fibro-adipose replacement)	clinical sign	variable (to adulthood)	severe, end-stage → transplant	subset	HP:0001644
Lactic acidosis	lab abnormality	neonatal–infantile	moderate–severe, episodic ↔ persistent	very common	HP:0003128
Leigh / Leigh-like encephalopathy	clinical sign	infantile–juvenile	progressive	common in neuro forms	HP:0002415
Ataxia	clinical sign	childhood–juvenile	slowly progressive	subset	HP:0001251
Optic atrophy	clinical sign	juvenile	progressive	subset	HP:0000648
Peripheral neuropathy	clinical sign	juvenile	progressive	subset	HP:0009830
Sensorineural hearing loss	clinical sign	childhood	progressive	subset	HP:0000407
Encephalopathy / developmental delay / hypotonia	clinical sign	infantile	severe	common in neuro forms	HP:0001263 / HP:0001252
Hepatic failure / dysfunction	clinical sign	neonatal	severe, often fatal	subset	HP:0001399 / HP:0001410
Intrauterine growth retardation	physical	prenatal	—	subset	HP:0001511

Quality-of-life impact. Severe across forms: infantile presentations are frequently fatal; survivors have major disability from cardiac failure, neurodegeneration (Leigh disease), blindness (optic atrophy), deafness, ataxia and neuropathy, with high dependency for daily functioning. No disease-specific QoL instrument (EQ-5D/SF-36) data exist; generic mitochondrial-disease QoL is markedly reduced. **[human clinical]** (P 17033963 P 25037205 P 30911037)

4. Genetic / Molecular Information

Causal gene. TSFM (HGNC:11701; NCBI Gene 10102; Ensembl ENSG00000123297; UniProt P43897; OMIM 604723), chromosome **12q14.1**. Encodes mitochondrial EF-Ts.

Reported pathogenic variants (all germline, autosomal recessive): - **c.997C>T, p.Arg333Trp** (a.k.a. R312W in shorter isoform numbering) — recurrent missense; most severe; homozygous in encephalomyopathy, HCM, and infantile liver failure (P 17033963 P 21741925). Also seen compound-het with p.Val119Leu (P 35071363).
- **c.944G>A, p.Cys315Tyr** and **c.856C>T, p.Gln286X** (nonsense) — Finnish, compound-het, juvenile Leigh/neuropathy/optic atrophy (P 25037205).
- **c.547G>A, p.Gly183Ser** — homozygous; encephalocardiomyopathy + sensorineural hearing loss (P 31267352).
- Novel compound-heterozygous variants causing dilated cardiomyopathy with fibro-adipose replacement

(P 30911037).

- A homozygous missense causing childhood ataxia + HCM (P 27677415).

Variant classification (ACMG/AMP): recurrent alleles are **Pathogenic/Likely Pathogenic** (functional rescue evidence PS3; recurrence/segregation; rarity in gnomAD). **Variant types:** predominantly **missense**, plus **nonsense (p.Q286X)** and **splice-site** alleles. **Allele frequencies:** very rare globally in gnomAD; p.Q286X enriched in Finland (~1:80 carriers). **Functional consequence: loss of function** via protein destabilization (reduced EF-Ts steady-state → secondary EF-Tu loss); no gain-of-function/dominant-negative mechanism. **[in vitro/computational]**

(P 17033963) P 25037205

Modifier genes. Not specifically identified for TSFM, but tissue-specific and nuclear-modifier influences are invoked to explain why identical genotypes yield divergent organ phenotypes; "unknown factors" are explicitly cited (P 27677415).

EF-Tu (TUFM) abundance is a candidate buffering modifier (P 30911037).

Epigenetic / chromosomal abnormalities. None described; COXPD3 is a single-gene point-mutation disorder without reported methylation changes or large structural/cytogenetic lesions.

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5. Environmental Information

Environmental factors: none causal. **Lifestyle factors:** not applicable to a monogenic infantile disease; avoidance of mitochondrial-toxic drugs and catabolic stressors is prudent management. **Infectious agents:** none causal; intercurrent infections can trigger metabolic decompensation. **[not applicable / inferred]**

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic TSFM mutation** → produces a **misfolded/unstable EF-Ts protein** (loss of function; R333W disrupts the subdomain structure and the EF-Ts dimerization/ EF-Tu-interaction interface — *demonstrated by molecular modeling and reduced steady-state protein*).
   (P 17033963)
2. → **reduced steady-state EF-Ts**, which **leads to secondary depletion of EF-Tu** (EF-Ts is EF-Tu's nucleotide-exchange factor; patient fibroblasts: EF-Ts −75%, EF-Tu −60%).
   (P 17033963)
3. → **impaired formation/recycling of the EF-Tu·GTP·aminoacyl-tRNA ternary complex**, which **results in defective delivery of aminoacyl-tRNAs to the mitoribosomal A-site** and **stalled translation elongation**. (*mechanistically inferred from EF-Ts/EF-Tu biochemistry; translation defect demonstrated*)
   (P 25037205)
4. → **decreased synthesis of the 13 mtDNA-encoded OXPHOS subunits**, which **leads to impaired assembly of respiratory-chain complexes I, III, IV and V** (complexes reduced 35–91%; complex II spared because it is fully nuclear-encoded).
   (P 17033963)
5. → **reduced oxidative phosphorylation / ATP output**, which **results in a shift to anaerobic glycolysis → lactate accumulation → lactic acidosis**.
   (P 21741925)
6. → **energy failure in high-demand post-mitotic tissues**, which **leads to cell dysfunction, apoptosis/necrosis, and tissue injury** — branching by organ:
7. **Heart** → hypertrophic or dilated cardiomyopathy, biventricular fibro-adipose replacement (P 30911037);
8. **CNS (basal ganglia, brainstem)** → Leigh-like necrotizing encephalopathy (P 25037205);

9. **Optic nerve / peripheral nerve** → optic atrophy, neuropathy;
10. **Liver** → hepatocellular failure
    (P 21741925);
11. **Cochlea** → sensorineural hearing loss
    (P 31267352).
12. **Compensatory branch (partially protective):** in some tissues, **EF-Tu up-regulation** + **induction of mitochondrial biogenesis** restores oxygen consumption, **explaining tissue-specific sparing** (e.g., fibroblasts normal despite affected myocardium).
    (P 30911037)

Supporting detail

- **Molecular pathways / biochemistry:** mitochondrial translation (KEGG hsa03010-analogous mitoribosome machinery); OXPHOS (KEGG hsa00190; Reactome "Respiratory electron transport"). GO biological processes: **mitochondrial translational elongation (GO:0070125)**, **oxidative phosphorylation (GO:0006119)**, **mitochondrial respiratory chain complex assembly (GO:0033108)**. GO molecular functions: **translation elongation factor activity (GO:0003746)**, **guanyl-nucleotide exchange factor activity (GO:0005085)**.
- **Cellular processes:** energy-deprivation-triggered apoptosis/necrosis; integrated stress / mitochondrial biogenesis response (inferred).
- **Protein dysfunction:** destabilization / loss of function at the EF-Ts–EF-Tu interface (PDB homologs of bacterial EF-Ts; AlphaFold model of P43897). **[computational]**
- **Metabolic changes:** elevated lactate, pyruvate, alanine; secondary TCA-cycle/ energy-metabolism disruption (CHEBI: lactate CHEBI:24996).
- **Immune involvement:** none (not an immune-mediated disease).
- **Tissue-damage mechanism:** bioenergetic failure/oxidative stress → cell death; cardiac fibro-adipose remodeling.
- **Molecular profiling:** patient myocardium/fibroblasts show reduced EF-Ts/EF-Tu, combined complex deficiency, and compensatory biogenesis transcriptional signatures
  (P 30911037)
  [in vitro]. No dedicated single-cell/spatial/CRISPR-screen datasets specific to TSFM.

Upstream vs downstream: mutation → protein instability (most upstream) → translation defect → OXPHOS assembly failure → ATP deficit/lactic acidosis → organ-specific cell death (most downstream). **Cell types (CL):** cardiomyocyte (CL:0000746), neuron (CL:0000540), hepatocyte (CL:0000182), skeletal muscle cell (CL:0000188). **Subcellular (GO-CC):** mitochondrion (GO:0005739), mitochondrial matrix (GO:0005759).

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7. Anatomical Structures Affected

- **Organ level (primary):** heart (UBERON:0000948); brain — basal ganglia (UBERON:0002420), brainstem (UBERON:0002298); optic nerve (UBERON:0000941); liver (UBERON:0002107); skeletal muscle (UBERON:0001134); peripheral nerve (UBERON:0001021); inner ear/cochlea (UBERON:0001844).
- **Body systems:** cardiovascular, central & peripheral nervous, hepatic/digestive, musculoskeletal, special sensory (visual, auditory).
- **Secondary involvement:** heart failure → systemic hypoperfusion; hepatic failure → coagulopathy; failure to thrive/IUGR.
- **Tissue/cell level:** cardiac muscle (cardiomyocytes, CL:0000746), CNS neurons/glia (CL:0000540), hepatocytes (CL:0000182), skeletal myocytes.
- **Subcellular:** mitochondrion / mitochondrial matrix (GO:0005739 / GO:0005759); biopsy shows mitochondrial hyperplasia/proliferation (P 35071363).
- **Localization / laterality:** CNS lesions are characteristically **bilateral and symmetric** (Leigh pattern). **[human clinical]**

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8. Temporal Development

- **Onset:** congenital/neonatal to juvenile. Most severe (R333W homozygous) → **neonatal–infantile**; hypomorphic genotypes → **childhood–adolescent** onset. Onset pattern ranges acute (neonatal lactic-acidotic crisis) to insidious/chronic (slowly progressive ataxia). (P 17033963)

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- **Progression:** generally **chronic-progressive**; cardiac disease can advance to end-stage requiring transplant; neuro forms show stepwise/progressive decline, sometimes with partially reversible neuroimaging changes
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- **Course pattern:** progressive, with episodic metabolic decompensations superimposed on a declining baseline.
- **Duration:** lifelong; infantile forms often fatal within the first months–years.
- **Critical periods:** neonatal/infancy is the window of greatest vulnerability and of greatest potential benefit from early supportive intervention. **[human clinical]**

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9. Inheritance and Population

- **Epidemiology:** no COXPD3-specific prevalence; it is ultra-rare (<1/1,000,000; only a few dozen families reported). Primary mitochondrial disease collectively affects **≥1 in 4,300**
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- **Inheritance: autosomal recessive** (biallelic).
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- **Penetrance:** effectively complete for biallelic pathogenic genotypes; **expressivity highly variable** (organ and severity differ even for identical genotypes)
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- **Genetic anticipation / germline mosaicism:** not reported / not applicable.

- **Founder effect: Finnish p.Q286X**, carrier frequency **~1:80**; notably **no homozygotes** among 35,000 controls or patients, consistent with lethality of that genotype (P 25037205).
- **Consanguinity:** major contributor to homozygous cases worldwide.
- **Carrier frequency:** high regionally (Finland); otherwise rare.
- **Demographics:** reported across populations (Finnish, Pakistani/Pakhtun, Mediterranean, Chinese, others); no strong sex bias (autosomal); age distribution skewed to infancy/childhood. **[human clinical]**

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10. Diagnostics

- **Laboratory:** elevated **blood & CSF lactate** (LOINC 2524-7), elevated pyruvate, **elevated plasma alanine**, lactic aciduria with increased TCA intermediates. (P 38571879)
- **Biopsy / biochemistry:** muscle or cardiac biopsy shows **combined reduction of respiratory-chain complexes I, III, IV with normal complex II**, plus **mitochondrial hyperplasia/proliferation** and sometimes COX-deficient fibres (P 35071363). P 17033963
  This **combined, complex-II-sparing** pattern is a key diagnostic signature.
- **Imaging:** brain MRI — **bilateral symmetric T2-hyperintense lesions of basal ganglia/brainstem** (Leigh pattern); **MR spectroscopy lactate peak** (P 38571879).
  Echocardiography for cardiomyopathy.
- **Electrophysiology:** ECG (cardiomyopathy/arrhythmia), nerve-conduction studies (neuropathy), BAER/audiometry (hearing loss).
- **Genetic testing (definitive): WES/WGS** or **NGS mitochondrial/nuclear multigene panels** identifying biallelic TSFM variants, with Sanger confirmation and parental segregation. **mtDNA testing is typically normal** (nuclear-gene disorder) and helps exclude mtDNA Leigh/mito-cardiomyopathy.

(P 25037205 P 30911037 P 35071363)

- **Omics diagnostics:** research-level proteomics/transcriptomics (reduced EF-Ts/EF-Tu; biogenesis signatures). No routine metabolomic/epigenomic test.
- **Clinical criteria / differential:** distinguish from other **mitochondrial translation-factor defects (GFM1/EF-G1, TUFM/EF-Tu, mt-aminoacyl-tRNA synthetases)**, **mtDNA-encoded Leigh syndrome**, **POLG disease**, and other combined-OXPHOS-deficiency genes. (P 35703069 P 22283595)
- **Screening:** no newborn screening; **carrier/cascade testing** and **prenatal/PGT** feasible once familial variants are known (relevant in Finnish founder and consanguineous families). **[human clinical]**

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11. Outcome / Prognosis

- **Survival/mortality: poor** overall; the recurrent severe allele causes **fatal infantile** encephalomyopathy, HCM, or neonatal liver failure (P 17033963 P 21741925). Hypomorphic genotypes allow survival into childhood/adolescence/adulthood (P 25037205 P 27677415).
- **Morbidity/disability:** high — cardiac failure, neurodegeneration, blindness (optic atrophy), deafness, ataxia, neuropathy; severe dependency.
- **Complications:** intractable heart failure, metabolic/lactic-acidotic crises, hepatic failure, respiratory failure, feeding difficulties.

- **Recovery:** limited; some neuroimaging changes partially reversible (PMID 31267352), but structural damage is generally irreversible. **Cardiac transplantation** rescued an adult with end-stage TSFM dilated cardiomyopathy (PMID 30911037).
- **Prognostic factors: genotype severity (residual EF-Ts function)**, age/organ of onset, and **degree of cardiac involvement** are the main determinants. No validated circulating prognostic biomarker specific to TSFM (GDF-15 is a general mitochondrial-disease marker). **[human clinical]**

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12. Treatment

No disease-specific or curative therapy exists. Management is **supportive, organ-directed, and empiric** (PMID 33105273, PMID 11282042).

- **Pharmacotherapy / "mito cocktail" (NCIT-relevant, empiric):** Coenzyme Q10/ubiquinone (CHEBI:16389), riboflavin (CHEBI:17015), thiamine (CHEBI:18385), L-carnitine (CHEBI:16347), **vitamin E** and **N-acetylcysteine** (strongest preclinical antioxidant evidence), B-vitamins; **L-arginine** for stroke-like episodes. **Vitamin C lacks efficacy**; **L-carnitine** use is nuanced (possible pro-atherogenic signal). Combinatorial **glucose** + **nicotinic acid** + **N-acetylcysteine** was synergistic in complex-I disease models. (PMID 33105273, PMID 33640978)
- **Pharmacogenomics:** avoid mitochondrial-toxic drugs (aminoglycosides, valproate, certain anaesthetics); no TSFM-specific PGx guidance.
- **Advanced therapeutics:** no approved gene, cell, RNA, or targeted therapy; none in trials specifically for TSFM. Gene replacement is conceptually attractive (EF-Ts cDNA rescues patient cells in vitro —

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but not clinically available.

- **Surgical/interventional: heart transplantation** for end-stage cardiomyopathy (NCIT heart transplantation)
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cardiac device/HF management; cochlear implantation for deafness (supportive).
- **Supportive/rehabilitative:** HF pharmacotherapy, anti-seizure medication, nutritional/feeding support, management of lactic acidosis; physical/occupational/speech therapy; low-vision and audiologic rehabilitation.
- **Treatment outcomes:** no validated response rates; evidence base weak owing to rarity and heterogeneity (no RCTs). **[human clinical / model organism]**

NCIT intervention terms: Coenzyme Q10 Therapy, Riboflavin, Thiamine, L-Carnitine, Vitamin E, N-Acetylcysteine, L-Arginine, Heart Transplantation, Supportive Care.

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13. Prevention

- **Primary prevention:** none (monogenic). **Genetic counseling** (25% recurrence risk per pregnancy for carrier couples), **carrier screening** (especially Finnish founder p.Q286X and consanguineous families), **prenatal diagnosis/PGT-M** once familial variants known (ACMG/NSGC frameworks, GeneReviews).
- **Secondary prevention:** early molecular diagnosis enables timely supportive care and surveillance (cardiac, neurologic, hepatic). No population newborn screening.
- **Tertiary prevention:** proactive cardiac monitoring/HF management, avoidance of catabolic stress and mitochondrial-toxic drugs, prompt treatment of intercurrent illness to prevent decompensation.
- **Immunization/public-health/environmental measures:** standard childhood vaccination to reduce infection-triggered crises; no disease-specific public-health intervention. **[human clinical / guideline-based]**

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** TSFM/EF-Ts is **deeply conserved from bacteria (E. coli *tsf*) through yeast to mammals**; the mitochondrial translation apparatus "resembles that found in prokaryotes" (P 17033963), reflecting endosymbiotic bacterial ancestry. Orthologs: **mouse *Tsfm*** (NCBI Gene 219138), rat, zebrafish, *Drosophila*, and bacterial *tsf*.
- **Natural disease (OMIA):** no naturally occurring TSFM-associated disease documented in companion animals or wildlife in the literature reviewed.
- **Comparative biology / evolutionary conservation:** strong conservation of the EF-Ts/EF-Tu nucleotide-exchange mechanism makes bacterial and yeast systems informative for mechanism (P 23727608).
- **Transmission / zoonosis:** not applicable (genetic disease). **[computational / comparative]**

15. Model Organisms

- **Model types used:** predominantly **in vitro human cellular models — patient-derived skin fibroblasts** and **cardiac/skeletal-muscle tissue**, with **wild-type TSFM (or TUFM) cDNA/retroviral complementation** as the gold-standard functional proof of causation (rescue of translation and OXPHOS assembly) (P 17033963 P 27677415) **[in vitro]**.
- **Yeast systems** have been used to study suppression of mitochondrial translation-elongation-factor defects, exploiting the conserved machinery (P 23727608) **[model organism]**.

- **Dedicated animal models: no purpose-built *Tsfm* knockout/knock-in mammalian model** was identified in the reviewed literature. A **constitutive *Tsfm* knockout is predicted to be embryonic-lethal** (consistent with the absence of surviving human null-homozygotes), so **conditional/tissue-specific** models would be required; iPSC-derived cardiomyocytes/organoids are a logical future platform.
- **Phenotype recapitulation / limitations:** patient-cell models faithfully reproduce the **molecular defect** (EF-Ts/EF-Tu loss, translation defect, combined complex deficiency) and allow rescue experiments, but **do not model organ-level pathology** (cardiomyopathy, Leigh-like neurodegeneration); this is the principal gap.
- **Resources:** MGI (mouse *Tsfm*), Cellosaurus/ATCC (patient fibroblast lines), Alliance of Genome Resources. **[in vitro / model organism]**

Summary Answer

Combined Oxidative Phosphorylation Deficiency Type 3 (COXPD3; MONDO:0012512, OMIM 610505) is a rare autosomal-recessive mitochondrial disease caused by biallelic mutations in *TSFM*, encoding mitochondrial translation elongation factor EF-Ts. Mutant EF-Ts is destabilized and secondarily depletes its partner EF-Tu, crippling mitochondrial translation elongation and the synthesis of the 13 mtDNA-encoded OXPHOS subunits, producing a **combined deficiency of respiratory-chain complexes I, III, IV and V (complex II spared)**, ATP failure and lactic acidosis. The clinical spectrum is broad and genotype-/organ-dependent — **hypertrophic or dilated cardiomyopathy, Leigh-like encephalopathy, ataxia, optic atrophy, peripheral neuropathy, sensorineural hearing loss, and infantile liver failure** — ranging from fatal neonatal multisystem disease (recurrent p.Arg333Trp allele) to survivable juvenile neurological forms. Diagnosis rests on lactic acidosis, a combined (complex-II-sparing) respiratory-chain enzyme defect on biopsy, Leigh-pattern MRI, and molecular confirmation of biallelic TSFM variants by WES/panel sequencing; a Finnish founder nonsense allele (p.Gln286X, carrier ~1:80) is notable. **No curative therapy exists** — management is supportive (mitochondrial cofactor/antioxidant "cocktail," heart-failure care, transplantation) — and prevention relies on genetic counseling and carrier/prenatal testing.

Supported hypotheses / conclusions

1. TSFM/EF-Ts loss of function is the sole, sufficient cause (rescue-confirmed). **Supported.**
2. The biochemical lesion is a combined, complex-II-sparing OXPHOS deficiency. **Supported.**
3. Phenotype severity tracks residual EF-Ts function; compensation explains tissue sparing. **Supported.**
4. Autosomal recessive with a Finnish founder allele. **Supported.**

Refuted / excluded

- Environmental, infectious, immune, epigenetic, or chromosomal primary causation. **Excluded** (no evidence; monogenic point-mutation disorder).
- mtDNA mutation as cause. **Excluded** (nuclear gene; mtDNA normal).

Limitations and future directions

- Ultra-rare: all data from individual case reports/small series; no prevalence, natural-history cohort, QoL, or RCT data specific to TSFM.
- No dedicated animal model; organ-level pathogenesis (why heart vs brain vs liver) is unresolved — modifier genes and tissue EF-Tu buffering are leading hypotheses.
- Therapeutic opportunity: gene replacement is biologically validated in vitro but clinically undeveloped; iPSC-cardiomyocyte/organoid models and targeted trials are needed.

Key references (PMID)

- 17033963 — Smeitink et al. 2006. Distinct phenotypes from an EF-Ts (TSFM) mutation; mechanism, rescue. **[landmark]**
- 21741925 — Vedrenne et al. 2012. TSFM R312W → infantile liver failure; same mutation, distinct syndromes.
- 25037205 — Ahola et al. 2014. Juvenile Leigh/neuropathy/optic atrophy; Finnish p.Q286X founder (1:80).
- 27677415 — Emperador et al. 2016. Childhood ataxia + HCM; cDNA rescue; variable phenotypes.
- 31267352 — Scala et al. 2019. p.G183S encephalocardiomyopathy + hearing loss; reversible imaging.

- 30911037 — Perli et al. 2019. Dilated cardiomyopathy, fibro-adipose replacement, transplant; compensation.
- 35071363 — Yang et al. 2021. HCM + lactic acidosis; compound het R333W/V119L; biopsy.
- 35703069 — Khan et al. 2022. Three mitochondrial elongation factors (TSFM/TUFM/GFM1).
- 33105273 — Barcelos et al. 2020. Mitochondrial medicine therapies, dosing, prevalence ≥1/4300.
- 33640978 — Guha et al. 2021. Combinatorial Glu+NA+NAC synergy in complex-I models.
- 23727608 — Montanari et al. 2013. Yeast model of mitochondrial translation factor suppression.
- 22283595 — Rahman 2012. Mitochondrial disease and epilepsy (classification context).
